



UITS

**UNIVERSITY OF INFORMATION
TECHNOLOGY AND SCIENCES**

UniQuery — “*Ask. Explore. Choose.*”

A Smart University Information and Admission Assistant

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System Modeling: Four Diagrams with Descriptions
Prepared from the UniQuery Project Proposal and Project Report

Project Overview

UniQuery is a smart, AI-powered university information and admission assistant. Its purpose is to help admission-seeking students search, evaluate, and compare university options using their academic profile, preferences, and available university information.

The project addresses three core challenges identified in the proposal: **complex eligibility assessment**, **time-consuming university comparison**, and **lack of real-time assistance**. The system follows the three-stage workflow **Ask, Explore, and Choose**.

The main features include a **Conversational AI Assistant**, **Smart Eligibility Matcher**, **Side-by-Side University Comparison**, and **Centralized Notice & Deadline Tracker**. The methodology moves from user profile input and AI/eligibility processing to data retrieval, program matching, comparison, and notice/deadline tracking.

Diagram	What it explains
Component Diagram	System components and their relationships.
Activity Diagram	Student workflow from input to final university choice.
Sequence Diagram	Chronological communication among system actors/services.
Fishbone Diagram	Root causes of university-selection difficulties and the UniQuery response.

1. Component Diagram

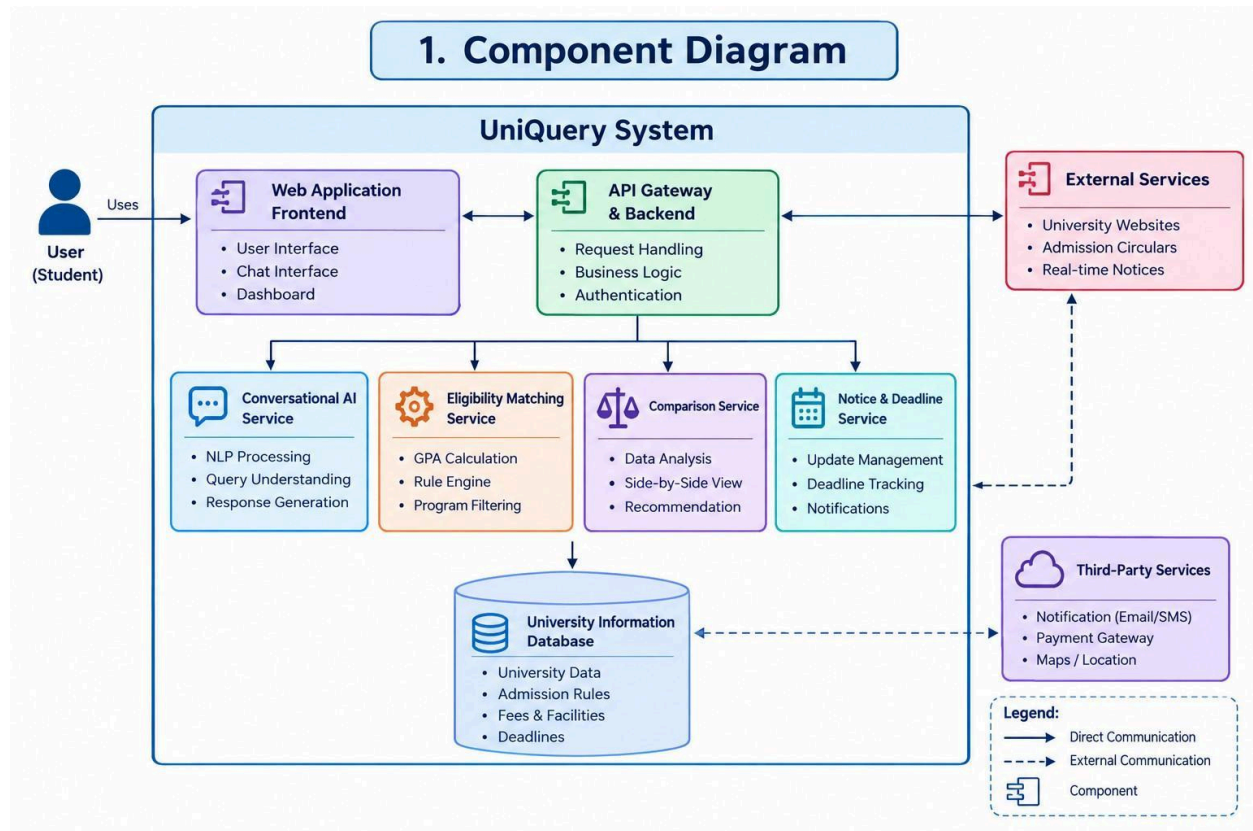


Figure 1: 1. Component Diagram

Description: The Component Diagram presents the major software components of the UniQuery system and their interactions. The diagram includes the Web Interface, AI Chatbot, University Information Database, Eligibility Matcher, Comparison Dashboard, and Notice & Deadline Tracker. The Web Interface is the main interaction point for students. The AI Chatbot handles natural-language admission queries, while the Eligibility Matcher evaluates student academic information against university-specific rules. The Comparison Dashboard supports side-by-side comparison of university information, and the Notice & Deadline Tracker manages important admission dates and application information.

Purpose: This diagram provides a high-level view of the UniQuery architecture and shows how its functional modules work together as an integrated university information and admission platform.

2. Activity Diagram

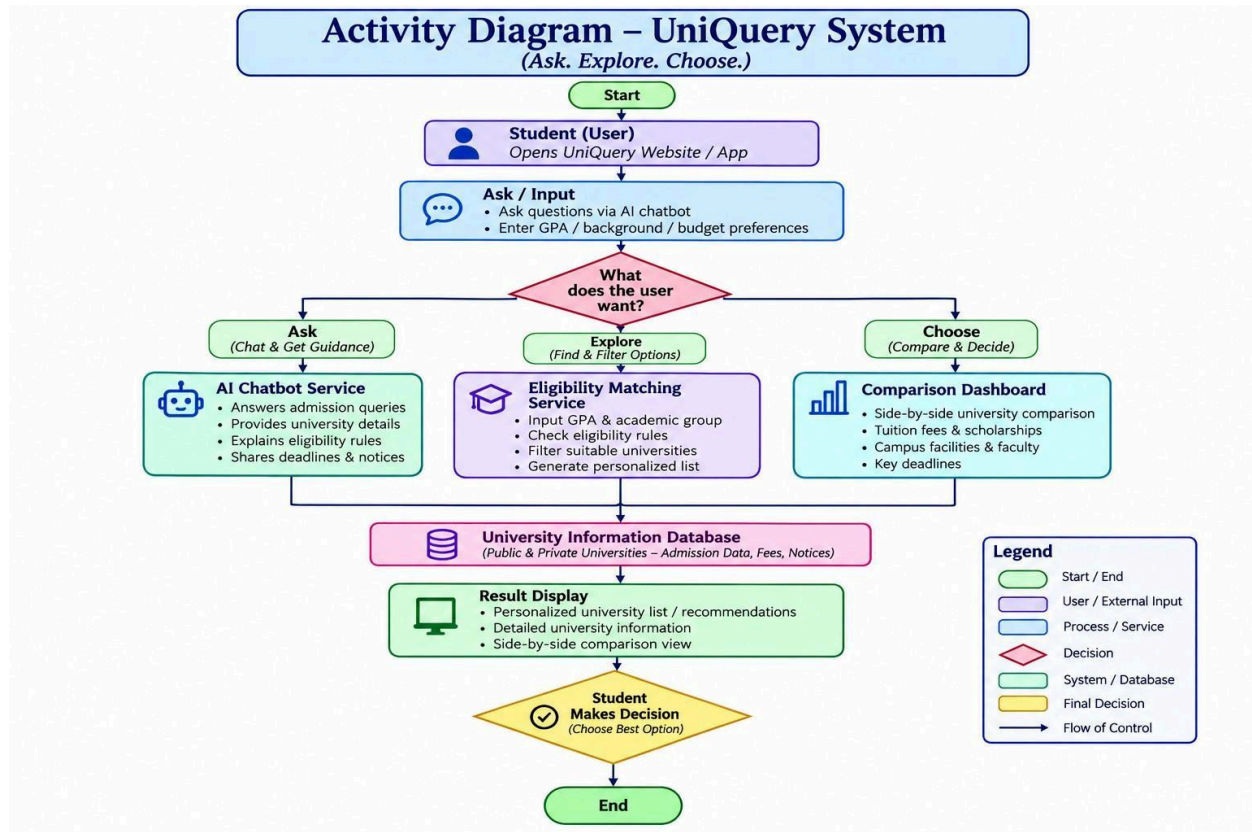


Figure 2: 2. Activity Diagram

Description: The Activity Diagram shows the step-by-step workflow of a student using UniQuery. The process starts when the student opens the platform and provides a query or academic profile. During the **Ask** stage, the student communicates with the AI chatbot. During **Explore**, the student provides information such as SSC/HSC GPA, academic group, subject information, and budget preferences. The eligibility engine checks the profile against stored university requirements and generates suitable programs. During **Choose**, the student compares shortlisted universities using factors such as tuition fees, scholarships, facilities, faculty information, and deadlines before selecting the preferred option.

Purpose: This diagram explains the complete user process from initial interaction and profile input through eligibility matching, exploration, comparison, and final decision-making.

3. Sequence Diagram

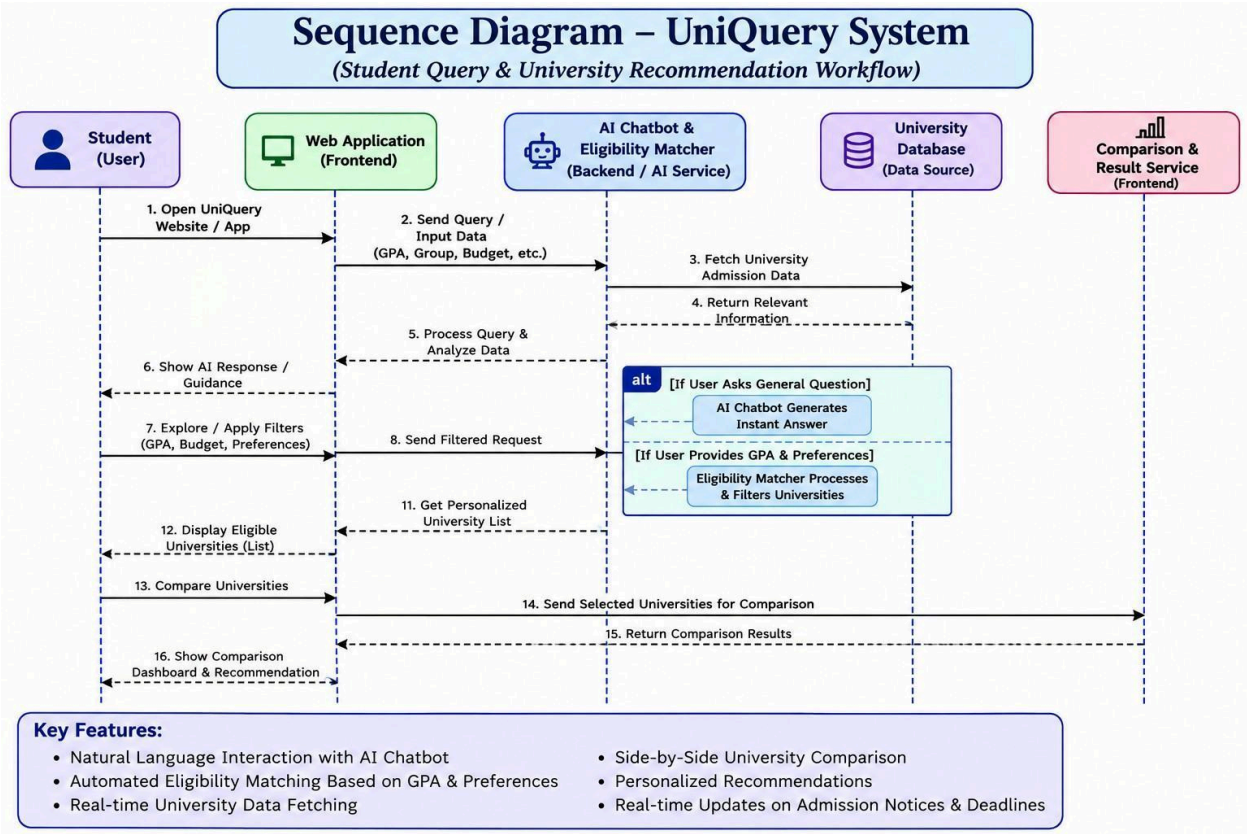


Figure 3: 3. Sequence Diagram

Description: The Sequence Diagram describes the chronological communication among the **Student**,

Web Application/Frontend, **AI Chatbot & Eligibility Matcher**, **University Database**, and

Comparison/Result Service. The student submits a natural-language query or academic profile through the frontend. The frontend forwards the request to the AI/backend service, which retrieves relevant information from the university database. The system processes the information and returns an answer or eligible university list. The student can then select universities for comparison, and the comparison service returns the comparative result to the dashboard.

Purpose: This diagram shows the order of messages and data flow between the student, frontend, AI/backend, database, and comparison/result services.

4. Fishbone Diagram

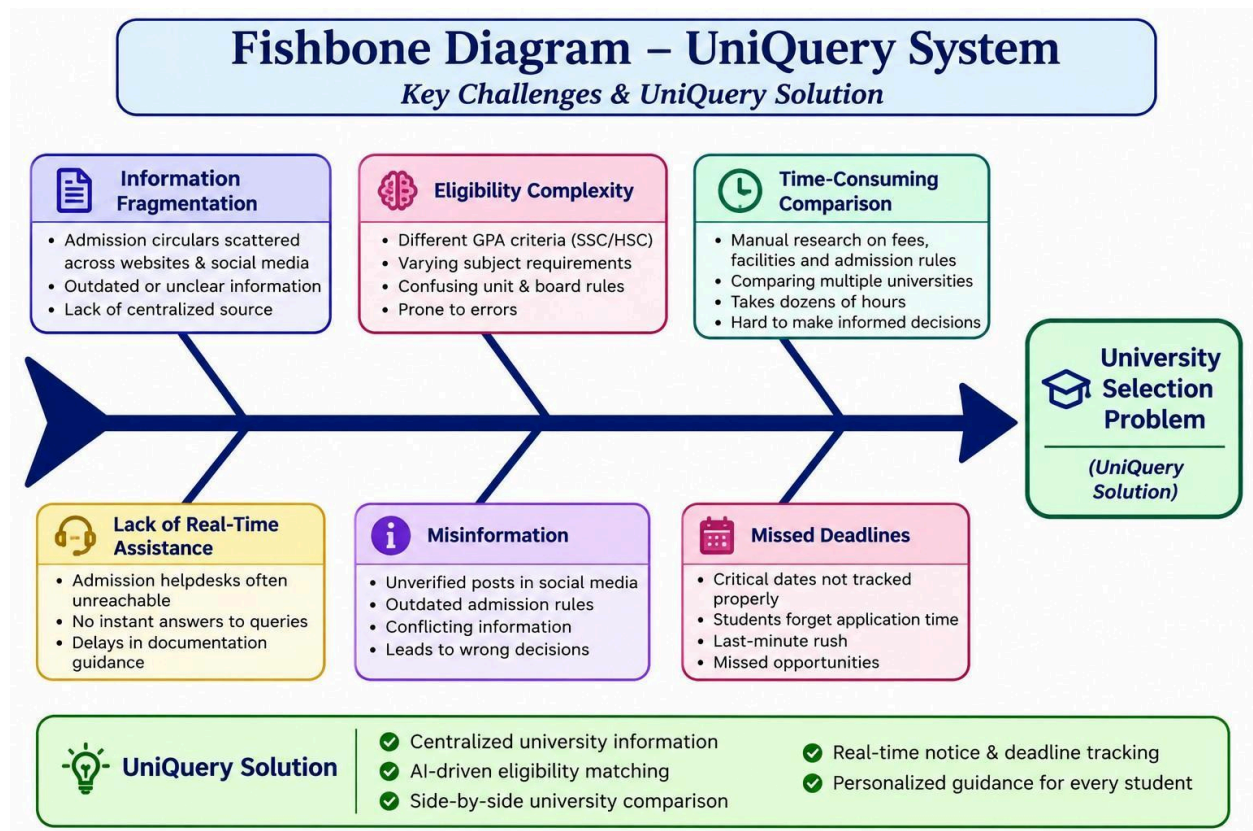


Figure 4: 4. Fishbone Diagram

Description: The Fishbone (Ishikawa) Diagram identifies the major causes behind difficulties in university selection. The central problem is the difficulty of selecting a suitable university and academic program. The main causes shown are **Information Fragmentation**, **Eligibility Complexity**, **Time-Consuming Comparison**, **Lack of Real-Time Assistance**, **Misinformation**, and **Missed Deadlines**. These causes reflect the challenges described in the proposal and report, such as scattered admission information, confusing eligibility rules, manual comparison, unavailable immediate guidance, unreliable information, and difficulty tracking critical dates.

Purpose: The diagram explains the root causes that motivate UniQuery and connects those problems to the system's centralized information, AI guidance, eligibility matching, comparison, and deadline-tracking features.